

4COSC002W MATHEMATICS FOR COMPUTING

Week 1 Seminar Tasks

Foundation Mathematics RECAP

READING

Chapters 1-10. Croft, T and Davison R (2016) *Foundation maths, 6th ed.* Harlow: Pearson.

1. CORE ARITHMETIC REFRESHER

THEORY

In computing and mathematics, we must work confidently with both **numbers** and **algebraic expressions**.

Key rules:

1. Order of Operations (BIDMAS)

- Brackets → Indices → Division/Multiplication → Addition/Subtraction.
- Ensures all calculations are done in the correct order – Left to Right.

2. Fractions and Negatives

- Fractions: add/subtract only with the same denominator; multiply/divide directly.
- Negative numbers: $(-) \times (-) = +$, $(+) \times (-) = -$.
- Absolute value $|x|$: always non-negative.

3. Simplifying Algebraic Expressions

- Collect like terms (combine coefficients of same variables).
- Expand brackets (use distributive law).
- Factorise (take out common factors).
- Work with algebraic fractions (use same rules as number fractions).

ESSENTIAL FORMULAE – Simplifying Expressions

Collecting Like Terms

$$ax + bx = (a + b)x$$

Expanding Brackets

Single Bracket (Distributive Law):

$$a(b + c) = ab + ac$$

$$a(b - c) = ab - ac$$

Double Brackets (FOIL):

$$(a + b)(c + d) = ac + ad + bc + bd$$

$$(a - b)(c + d) = ac + ad - bc - bd$$

$$(a - b)(c - d) = ac - ad - bc + bd$$

Special Quadratic Expansions:

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

$$(x + a)^2 = x^2 + 2ax + a^2$$

$$(x - a)^2 = x^2 - 2ax + a^2$$

$$(x - a)(x + a) = x^2 - a^2$$

Factorisation

$$ax + ay = a(x + y)$$

$$x^2 + (a + b)x + ab = (x + a)(x + b)$$

$$x^2 - a^2 = (x - a)(x + a)$$

Algebraic Fractions

$$\frac{a}{m} + \frac{b}{n} = \frac{an + bm}{mn}$$

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

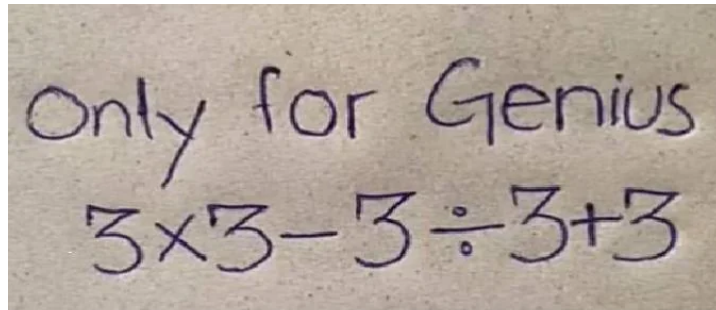
$$\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$$

$$\frac{ax}{bx} = \frac{a}{b}, (x \neq 0)$$

TASKS

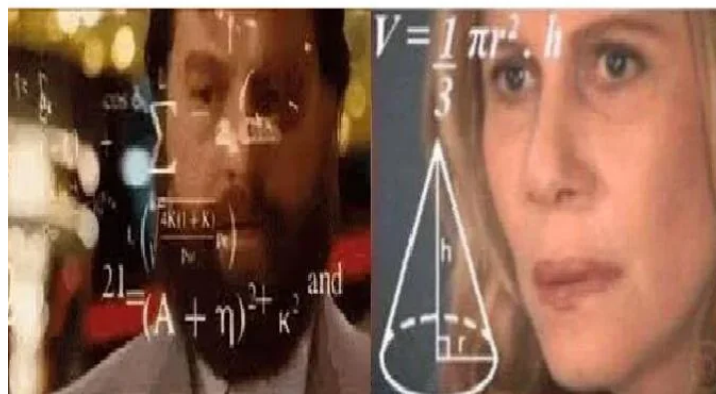
Try to solve the viral maths problems:

(1)

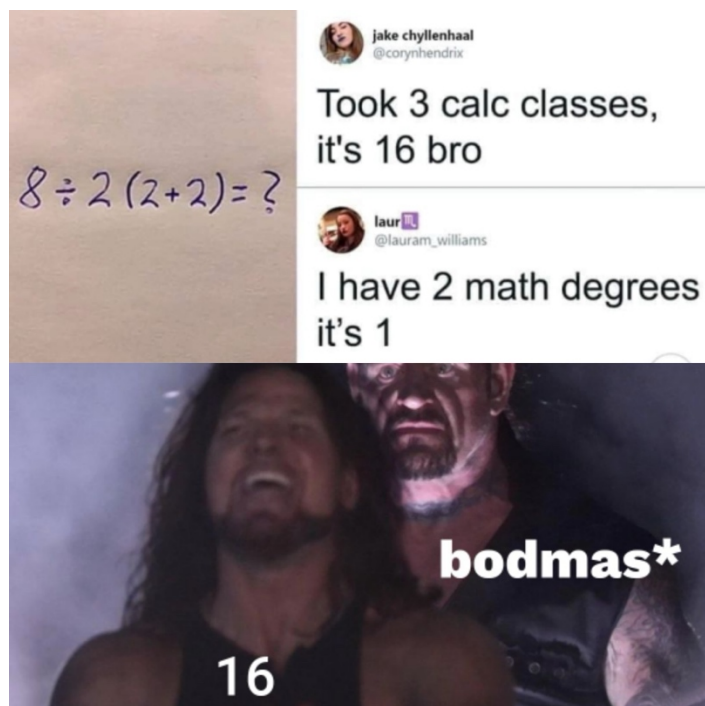


Only for Genius
 $3 \times 3 - 3 \div 3 + 3$

The Comment Section:



(2)



$8 \div 2 (2+2) = ?$

jake chyllenhaal
@corymhendrix
Took 3 calc classes,
it's 16 bro

laur
@lauram_williams
I have 2 math degrees
it's 1

bodmas*
16

Task 1 (calculate, multi-step BIDMAS) IN-CLASS ACTIVITY

Calculate:

$$2 + (3^2 - 4) \times 2 - 12 \div 3$$

Task 2 (calculate, fractions + brackets) IN-CLASS ACTIVITY

Calculate:

$$\frac{(12 - 4)^2}{8} - |-3| + 5$$

Task 3 (simplify, collecting like terms)

Simplify:

$$4x + 7y - 3x + 2y - 5$$

Task 4 (simplify, expand and collect)

Simplify:

$$2(x + 3) - (3x - 4)$$

Task 5 (simplify, expand double brackets)

Simplify:

$$(x + 2)(x - 5)$$

Task 6 (simplify, factorisation)

Simplify:

$$6x^2 + 9x$$

Task 7 (simplify, algebraic fraction)

Simplify:

$$\frac{4x - 6}{2x} + \frac{x}{2}$$

Task 8 (challenge – multiple operations)

Simplify:

$$\frac{2(x+1)(x-1)}{x} - (x^2 - 1)$$

2. ALGEBRA ESSENTIALS

THEORY

Algebra is about equations with unknowns (like x or y). The aim is to **isolate the unknown** by performing the same operation on both sides of the equation so the balance is preserved.

We will look at three key types:

1. **Linear equations** – only x , no powers.
2. **Equations with fractions** – require clearing denominators.
3. **Quadratic equations** – include x^2 , usually with two solutions.

TASKS

Linear Equations

A **linear equation** has the form $ax + b = c$.

Steps:

1. Expand brackets.
2. Collect like terms.
3. Rearrange: move x -terms to one side, constants to the other.
4. Divide to solve for x .

Task 1 (basic)

Solve:

$$2x + 7 = 15$$

Task 2 (with brackets)

Solve:

$$3(x - 5) + 2 = 10$$

Task 3 (harder, negatives both sides) **IN-CLASS ACTIVITY**

Solve:

$$2(3x - 4) - (x + 5) = 3(2x - 1) - 10$$

Equations with Fractions

When fractions are involved, first find the **lowest common denominator (LCD)** and multiply through to remove the denominators.

Task 4 (simple fractions)

Solve:

$$\frac{x}{3} + 2 = \frac{2x}{5}$$

Task 5 (two fractions)

Solve:

$$\frac{x+1}{4} - \frac{x-2}{3} = 1$$

Task 6 (harder fractions, both sides) **IN-CLASS ACTIVITY**

Solve:

$$\frac{2x-1}{5} = \frac{x+2}{3} - 1$$

Quadratic Equations

1. Definition

A quadratic equation has the form:

$$ax^2 + bx + c = 0, a \neq 0$$

The graph $y = ax^2 + bx + c$ is a parabola.

Solving means finding the **roots** (the x -values where the curve meets the x -axis).

2. Number of Solutions

Depends on the **discriminant** $D = b^2 - 4ac$:

- $D > 0$: two real solutions,
- $D = 0$: one repeated solution,
- $D < 0$: no real solutions (complex).

3. Solving Methods

- **Factorisation:**

The quadratic $ax^2 + bx + c$ can sometimes be expressed as the product of two linear factors:

$$(px + q)(rx + s) = 0$$

This relies on finding two numbers whose product gives ac and whose sum gives b . Once factorised, the **Zero Product Law** ($AB = 0 \Rightarrow A = 0$ or $B = 0$) is applied to find the solutions.

- **Completing the Square:**

Any quadratic can be rewritten in the form:

$$\left(x + \frac{b}{2a}\right)^2 = \text{constant}$$

This is done by adding and subtracting the square of half the coefficient of x . The method transforms the quadratic into a perfect square plus or minus a constant, which can then be solved by taking square roots. It also reveals the symmetry of the parabola and its turning point at $x = -\frac{b}{2a}$.

- **Quadratic Formula:**

Derived from completing the square, the general solution is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

This works for all quadratics. The value inside the square root, the **discriminant** $D = b^2 - 4ac$, determines the nature of the solutions (two, one, or none in the real numbers).

Task 7 (quadratic — factorisation) IN-CLASS ACTIVITY

Solve:

$$x^2 - 5x + 6 = 0$$

Task 8 (quadratic — completing the square)

Solve:

$$x^2 + 6x + 5 = 0$$

Task 9 (quadratic — completing the square, with coefficient)

Solve:

$$2x^2 + 8x + 3 = 0$$

Task 10 (quadratic — quadratic formula)

Solve:

$$3x^2 - 2x - 1 = 0$$

Task 11 (Quadratic Equation. UI Panel Sizing under Area Constraint) IN-CLASS ACTIVITY**Problem (for students):**

A UI panel must have a **fixed pixel area of 57,600**. For layout reasons, the panel's **width must exceed its height by exactly 48 px** (e.g., to align with a sidebar gutter). The panel is rectangular and not resizable except by changing width and height while keeping those two requirements. Find the panel's dimensions (width and height in pixels).

Index notes (CS terms):

- **Pixel:** smallest unit of a digital image or UI element.
- **UI panel:** rectangular region in a user interface layout.
- **Area (pixels²):** product of width and height, determines **resolution** and layout space.
- **Constraint-based layout:** widely used in **responsive UI design** (Android, iOS, web CSS grids).
- **Quadratic relation:** arises naturally in problems where one dimension depends on another and area is fixed.

3. SYSTEMS OF EQUATIONS

THEORY

A **system of equations** means solving two or more equations *together*, to find values of variables that satisfy all equations at once.

We usually deal with **two equations in two unknowns**:

$$\begin{cases} ax + by = c \\ dx + ey = f \end{cases}$$

Graphically, the solution is the **intersection point** of two lines.

Methods to solve:

1. **Substitution** – solve one equation for a variable, substitute into the other.
2. **Elimination** – combine equations to cancel one variable.
3. **Matrix methods** – introduced later in the module.

Systems can have:

- **One solution** (lines cross once).
- **No solution** (lines parallel).
- **Infinite solutions** (lines coincide).

TASKS

Task 1 (simple – substitution)

Solve:

$$\begin{cases} x + y = 10 \\ x - y = 4 \end{cases}$$

Task 2 (medium – elimination)

Solve:

$$\begin{cases} 2x + 3y = 12 \\ 3x - 2y = -1 \end{cases}$$

Task 3 (harder – fractions)

Solve:

$$\begin{cases} \frac{x}{2} + y = 5 \\ x - \frac{y}{2} = 3 \end{cases}$$

Task 4 (special case – no solution)

Solve:

$$\begin{cases} x + y = 4 \\ 2x + 2y = 10 \end{cases}$$

Task 5 (System of Equations. Cloud Billing Blocks) **IN-CLASS ACTIVITY**

A cloud platform bills in blocks, not per unit.

- Data transfer is charged in **blocks of 80 MB** (any partial block is billed as a full one).
- CPU time is charged in **blocks of 12 seconds** (again, partial → full block).

In one run, a job moves **310 MB** and runs for **29 s**; the bill for these two items together is **£9.40**.

In a second run, a job moves **670 MB** and runs for **41 s**; the bill for these two items is **£16.30**.

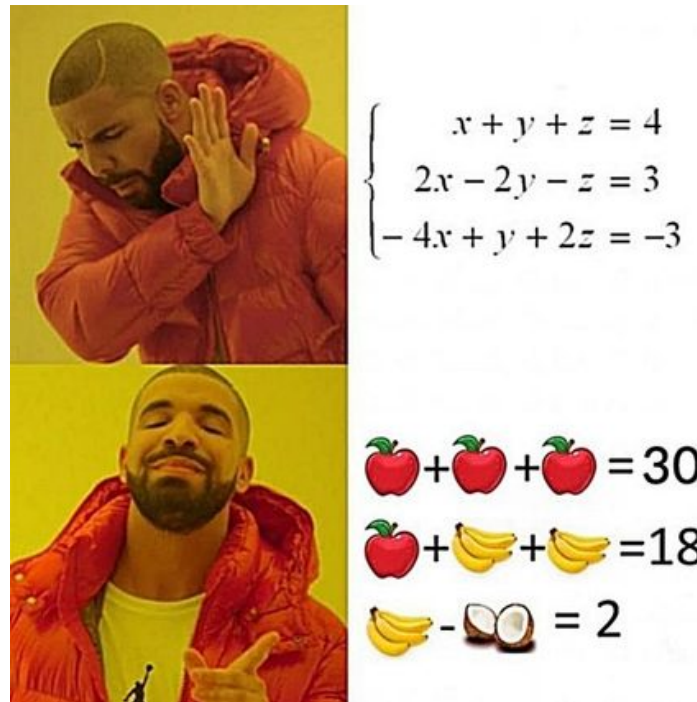
Assuming the platform uses a fixed price per data block and a fixed price per time block (the same across both runs), determine those two prices.

Notes (CS terms):

- **Data transfer (MB):** volume of information sent across a network.
- **CPU time (s):** processing capacity consumed by tasks.

- **Block billing:** common in **cloud services** (AWS, Azure, GCP), charging in chunks not per unit.
- **Cost modelling:** mapping resource use to linear equations is key in **cloud computing economics**.

Task 6 (Meme Task)



4. BASIC INEQUALITIES

THEORY

An **inequality** compares two expressions and shows a range of possible values for the variable. The symbols are:

- $<$ (less than), $\leq$ (less than or equal to)
- $>$ (greater than), $\geq$ (greater than or equal to)

We solve inequalities almost like equations: expand, collect terms, simplify, isolate the variable.

Important: when multiplying or dividing both sides by a **negative number**, the inequality **reverses direction**.

Inequalities usually give **solution sets** (many possible values), not just one solution. These can be shown:

- On a **number line** (with open/closed circles).
- As an **interval** (e.g., $x > 2 \Rightarrow (2, \infty)$).

TASKS

Task 1 (multi-step linear inequality with brackets)

Solve:

$$2(x - 3) + 5 \leq 3x - 7$$

Task 2 (fractions and negatives)

Solve:

$$\frac{2x - 3}{4} - \frac{x + 1}{3} > 1$$

Task 3 (compound inequality with brackets and negatives)

Solve:

$$-2 \leq 3x - 4 < 11$$

Task 4 (quadratic inequality)

Solve:

$$(x - 1)(x - 4) \leq 6$$

5. EXPONENTS AND LOGARITHMS

THEORY

Exponents (Powers)

- a^n means a multiplied by itself n times. Example: $2^4 = 2 \times 2 \times 2 \times 2 = 16$.
- Laws of exponents:
 - $a^m \cdot a^n = a^{m+n}$
 - $\frac{a^m}{a^n} = a^{m-n}$
 - $(a^m)^n = a^{mn}$
 - $a^{-n} = \frac{1}{a^n}$

Exponents are used to model growth (population, computing complexity, compound interest).

Logarithms

A logarithm is the **inverse of an exponent**.

$$\log_a b = c \Leftrightarrow a^c = b$$

- Example: $\log_2 8 = 3$ because $2^3 = 8$.
- Common bases:
 - $\log_{10}$ = “common log” (scientific calculators).
 - $\ln(x) = \log_e x$ = “natural log”, base $e \approx 2.718$.

Laws of logarithms (mirroring exponent laws):

- $\log_a (xy) = \log_a x + \log_a y$
- $\log_a (x/y) = \log_a x - \log_a y$
- $\log_a (x^n) = n \log_a x$

TASKS

Task 1 (simple exponent)

Simplify:

$$2^3 \cdot 2^4$$

Task 2 (exponent with negative power)

Simplify:

$$5^{-2}$$

Task 3 (simple logarithm) **IN-CLASS ACTIVITY**

Evaluate:

$$\log_2 16$$

Task 4 (logarithm property) **IN-CLASS ACTIVITY**

Simplify:

$$\log_{10} 1000 - \log_{10} 10$$

Task 5 (harder – solve exponent equation)

Solve for x :

$$3^x = 81$$

Task 6 (harder – solve with logs)

Solve for x :

$$2^x = 40$$

Task 7 (challenging – logarithmic equation)

Solve for x :

$$\log_3 (x) + \log_3 (x - 2) = 2$$

6. FACTORIALS AND COMBINATORICS

THEORY

Factorials

A **factorial** (written as $n!$) means multiplying all whole numbers from n down to 1:

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 2 \times 1$$

- Example: $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.
- Special case: by definition, $0! = 1$. This makes formulas work consistently.
- Factorials grow **very quickly**:
 - $3! = 6$
 - $6! = 720$
 - $10! = 3,628,800$

Factorials are used for **counting arrangements**. For example, the number of different ways to arrange 5 books on a shelf is $5!$.

Permutations (order matters)

A **permutation** counts how many ways we can arrange objects **when the order matters**.

- If we arrange *all* n objects:

$$P(n) = n!$$

- If we only arrange r objects out of n :

$$P(n, r) = \frac{n!}{(n - r)!}$$

This formula works because:

- Start with $n!$ (all arrangements of n),
- Then divide out the arrangements of the unused $(n - r)$ objects.

Example intuition: If 3 people are chosen to be first, second, and third in a race out of 10, the order **first vs second vs third** matters → permutations.

Combinations (order does not matter)

A **combination** counts how many ways to choose objects **when the order does not matter**.

- Formula:

$$C(n, r) = \binom{n}{r} = \frac{n!}{r! (n - r)!}$$

Why divide by $r!$? Because in combinations, the same r objects can be arranged in $r!$ different orders — but in combinations these orders are considered identical, so we divide them out.

Example intuition: If 3 students are chosen for a group project from 10, we don't care about their order $\rightarrow$ combinations.

TASKS

Task 1 (simple factorial) **IN-CLASS ACTIVITY**

Compute:

$$5!$$

Task 2 (permutations – full set)

How many different ways can 6 people be arranged in a line?

Task 3 (permutations – choosing r objects) **IN-CLASS ACTIVITY**

How many different 3-digit numbers can be formed from the digits 1,2,3,4 without repetition?

Task 4 (combinations – simple)

From 10 students, how many ways can we choose a team of 3?